# Jay Yu-Che Tsai

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Portfolio: jay-tsai-portfolio.vercel.app

## EDUCATION

**University of Southern California (USC)** — Los Angeles, CA
M.S. Computer Science (Game Development), Aug. 2024 – May 2026 | GPA: **3.82**
**National Yang Ming Chiao Tung University (NYCU)** — Hsinchu, Taiwan
*Bachelor of Science in Electronics and Electrical Engineering*
Sep. 2018 – Jun. 2022 | GPA: **3.5**

## GAME DESIGN PROJECTS

**The Unrealtor** | Gameplay Programmer / Systems Designer Jun. 2025 – May 2026

- Designed a cooperative puzzle experience built around an accessible perspective-based mechanic that encouraged communication and collaboration.
- Designed intuitive puzzle progression that gradually introduced new mechanics.
- Designed gameplay systems supporting cooperative communication and puzzle solving.
- Rapidly prototyped and iterated on original gameplay mechanics through structured playtesting and player feedback.
- Collaborated with programmers and artists to refine gameplay features through iterative development.
- Created gameplay documentation and design specifications to support implementation and future iteration.

**Whispers From the Deep** | Technical Game Designer / Gameplay Programmer Jan. 2025 – May 2025

- Designed, prototyped, and balanced the core spell-casting gameplay system, including player abilities, progression, and encounter pacing.
- Refined player controls and tutorial flow through weekly playtesting and rapid iteration.
- Balanced gameplay systems and progression using structured playtest feedback.
- Created gameplay documentation and feature specifications to support iterative development.

**Typomancer** | Technical Game Designer Sept. 2025 – Dec. 2025

- Designed player progression, class systems, and gameplay mechanics for distinct cooperative playstyles.
- Balanced gameplay systems and player progression through iterative playtesting and feedback.
- Created gameplay documentation and feature specifications to support iterative development.

**Mixed Reality Game** | Gameplay Programmer Spring 2022

- Designed and prototyped a cooperative puzzle experience using Unreal Engine and OSC-based multiplayer interaction.
- Implemented gameplay systems and gameplay UI supporting cooperative play.
- Rapidly prototyped cooperative gameplay interactions by integrating Unreal Engine, OSC communication, and Arduino hardware.

**Robotics Projects** | Software Engineer 2021–2022

- Designed autonomous robots using Arduino and Raspberry Pi (C++)
- Applied computer vision (YOLO) and voice recognition (PICOVOICE)
- Placed 2nd in the NYCU Robot Hockey Contest
- Integrated multiple sensors (IR, touch, light) to support autonomous decision-making

## Technical Skills

**Languages :** c++, c# Python, Java
**Engines:** Unreal Engine 5, Unity
**Gameplay:** Gameplay Systems, Gameplay Mechanics, Rapid Prototyping, Feature Design, Puzzle Design, Player Progression, Game Balance, Gameplay Documentation, Playtesting